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Detection of Water Leaks in Urban Distribution Networks Using a CNN-BiLSTM Architecture with Self-Attention

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ABSTRACT This paper presents an end-to-end deep learning framework for automatic leak detection in urban water distribution networks using multivariate sensor time series. The dataset consists of 5,590 samples collected from ten sensors over a three-week monitoring period, including 774 labelled leak events (13.8%). The proposed pipeline integrates domain-informed feature engineering, sliding-window sequence construction ($W = 20$, $S = 5$), and a temporal train/calibration/test split to prevent data leakage.

We propose a hybrid deep learning architecture combining Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and a self-attention mechanism. The CNN layers capture local temporal patterns, the BiLSTM models long-range bidirectional dependencies, and the attention mechanism highlights the most informative time steps. The model is trained using binary cross-entropy and evaluated with imbalance-aware metrics, including AUC-PR and calibrated decision thresholds.

Experimental results demonstrate strong performance, achieving $\text{AUC-ROC} = 0.9412$ and $\text{AUC-PR} = 0.9583$. At the selected operating threshold $\tau = 0.0333$, the model attains precision = 0.9647, recall = 0.9011, and F1-score = 0.9318, while maintaining a low false alarm rate of 0.0231. These results highlight a well-balanced trade-off between detection sensitivity and false alarms, indicating that the proposed approach is suitable for reliable deployment in real-world smart water monitoring systems.

KEY-WORDS Water Leak Detection, Deep Learning, CNN-BiLSTM, Self-Attention, Multivariate Time Series, Sliding Window, Imbalanced Classification, Threshold Calibration, Smart Water Networks;

I. INTRODUCTION

Water distribution networks are critical infrastructures for urban environments, ensuring the continuous delivery of potable water to residential, industrial, and commercial users. However, these systems are highly susceptible to leaks due to aging infrastructure, pressure fluctuations, and environmental factors. Undetected leaks can lead to significant water losses, increased operational costs, reduced service efficiency, and long-term structural damage. Consequently, the development of reliable and automated leak detection systems has become an important research and industrial challenge.

Traditional leak detection methods typically rely on manual inspection, acoustic sensing, or rule-based thresholding techniques applied to pressure and flow measurements.

While these approaches can be effective in controlled scenarios, they often lack scalability and adaptability in complex urban networks. Moreover, such methods struggle to capture the temporal dynamics and nonlinear patterns associated with leak events, particularly in the presence of noise and varying operating conditions.

With the increasing deployment of sensor networks and Internet of Things (IoT) technologies in smart water systems, large volumes of multivariate time-series data are now available. This has opened the door for data-driven approaches, particularly deep learning methods, which have demonstrated strong capabilities in modeling sequential data. Architectures such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have been widely used for time-series classification tasks due to their ability to capture local and long-range dependencies, respec-

tively. However, individual models may not fully exploit the complementary strengths of these approaches.

In this work, we propose a hybrid deep learning architecture that combines CNN, Bidirectional LSTM (BiLSTM), and a self-attention mechanism for automatic water leak detection. The CNN component extracts local temporal features from sensor signals, while the BiLSTM captures bidirectional dependencies across time. The integration of a self-attention layer allows the model to focus on the most informative segments of the input sequence, improving interpretability and detection performance.

Beyond the model architecture, we design a complete end-to-end pipeline tailored for real-world deployment. This includes domain-informed feature engineering, sliding-window sequence construction, and a strict temporal separation between training, calibration, and test sets to prevent data leakage. Furthermore, we address the inherent class imbalance in leak detection by adopting evaluation metrics such as AUC-PR and by performing threshold calibration to control the trade-off between detection sensitivity and false alarm rate.

Experimental results on a real-world dataset demonstrate that the proposed approach achieves strong performance, with high AUC-ROC and AUC-PR scores, as well as a favorable balance between precision and recall. In particular, the model maintains a low false alarm rate while achieving high leak detection accuracy, making it suitable for practical monitoring systems where both reliability and responsiveness are essential.

The main contributions of this paper can be summarized as follows:

- A hybrid CNN-BiLSTM architecture enhanced with a self-attention mechanism for multivariate time-series leak detection.
- A robust data processing pipeline including feature engineering, sliding-window transformation, and leakage-free temporal splitting.
- An imbalance-aware evaluation framework incorporating AUC-PR and threshold calibration for operational decision-making.
- Extensive experimental validation demonstrating strong detection performance with a balanced trade-off between recall and false alarm rate.

II. RELATED WORK

Automatic leak detection in water distribution networks has been widely studied, with approaches ranging from traditional signal processing techniques to modern data-driven machine learning models. Existing methods can be broadly categorized into three groups: physics-based approaches, classical machine learning methods, and deep learning-based models.

II.1. Traditional and Physics-Based Methods

Early work in leak detection primarily relied on hydraulic modeling and signal processing techniques. Methods based on pressure residual analysis, acoustic sensing, and flow balance estimation have been widely used in practice. These approaches typically depend on domain-specific knowledge and require accurate physical models of the water network. While effective in controlled environments, they often struggle in real-world scenarios due to noise, uncertainty, and the complexity of urban infrastructure systems.

II.2. Machine Learning Approaches

With the increasing availability of sensor data, machine learning techniques have been introduced for leak detection tasks. Classical models such as Support Vector Machines (SVM), Random Forests, and Gradient Boosting have been applied to classify normal and anomalous behavior based on extracted features. These approaches improve scalability compared to rule-based systems but often require extensive feature engineering and may fail to capture temporal dependencies inherent in time-series data.

II.3. Deep Learning for Time-Series Leak Detection

Recent advances in deep learning have enabled more effective modeling of multivariate time-series data. Convolutional Neural Networks (CNN) have been used to capture local temporal patterns, while Recurrent Neural Networks (RNN), particularly Long Short-Term Memory (LSTM) networks, are capable of modeling long-term dependencies.

Several studies have explored hybrid architectures combining CNN and LSTM to leverage both spatial and temporal feature extraction. These models have demonstrated improved performance compared to standalone approaches, especially in complex monitoring environments.

More recently, attention mechanisms have been introduced to enhance sequence modeling by allowing models to focus on the most relevant time steps. Attention has been successfully applied in various time-series applications, improving both performance and interpretability. However, its application to water leak detection remains relatively limited.

II.4. Limitations of Existing Work

Despite these advances, several limitations remain. Many existing studies do not adequately address class imbalance, which is a critical challenge in leak detection where leak events are relatively rare. Furthermore, evaluation is often limited to accuracy-based metrics, which may not reflect real-world operational requirements such as minimizing false alarms.

In addition, some approaches do not ensure strict separation between training and test data in temporal settings, potentially leading to data leakage and over-optimistic performance estimates.

II.5. Proposed Approach

To address these limitations, this paper proposes a hybrid CNN-BiLSTM architecture augmented with a self-attention mechanism for multivariate time-series leak detection. In contrast to prior work, the proposed approach integrates:

- A combination of CNN and BiLSTM to capture both local and long-range temporal patterns,
- A self-attention mechanism to identify the most informative time steps,
- A leakage-free temporal data splitting strategy,
- An imbalance-aware evaluation framework using AUC-PR and threshold calibration.

This combination enables robust and reliable leak detection while maintaining a balanced trade-off between detection sensitivity and false alarm rate.

III. DATASET AND EXPLORATORY DATA ANALYSIS

III.1. Dataset Description

The dataset used in this study consists of multivariate time-series data collected from an urban water distribution network over a monitoring period of approximately three weeks (January 2024). The data includes measurements from ten sensors capturing key hydraulic variables such as pressure and flow rate, along with environmental variables including temperature.

After preprocessing and removal of anomalous burst events, the dataset contains a total of 5,590 samples, among which 774 correspond to leak events, representing approximately 13.8% of the data. This class imbalance reflects real-world conditions, where leak occurrences are relatively rare compared to normal system operation.

Each observation is associated with a binary label indicating the presence or absence of a leak. To enable supervised learning, the data is later transformed into fixed-length sequences using a sliding-window approach.

III.2. Distribution Analysis

To better understand the characteristics of the data, we first analyze the distribution of key variables for normal and leak conditions.

As shown in Figure 1, pressure values during leak events tend to be lower compared to normal operating conditions. This shift reflects the physical behavior of the system, where leaks lead to pressure drops in the network.

Figure 2 shows that flow rate values are generally higher and more dispersed during leak conditions. This behavior is consistent with the expectation that leaks introduce additional flow demand, increasing the variability of the signal.

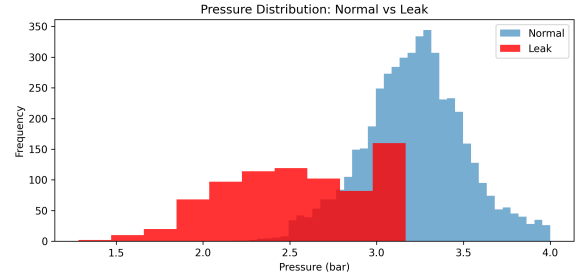


Figure 1: Pressure distribution under normal and leak conditions.

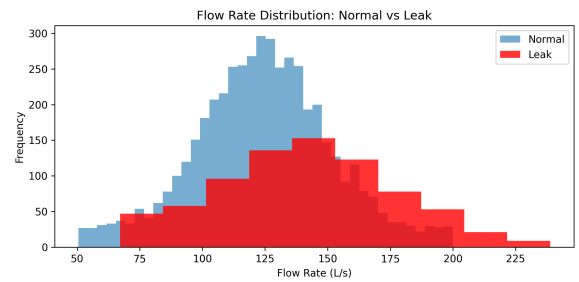


Figure 2: Flow rate distribution under normal and leak conditions.

III.3. Temporal Behavior of Leak Events

To analyze the temporal dynamics of the system, we examine sensor-level time-series data with annotated leak events.

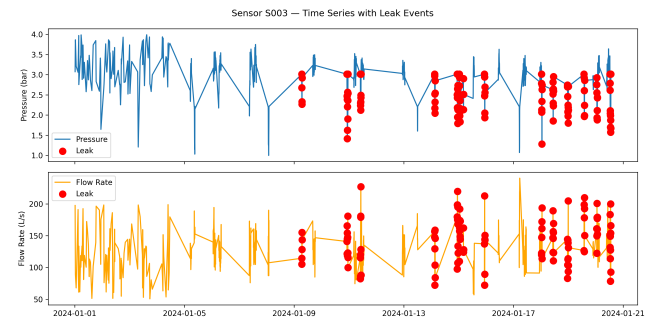


Figure 3: Time-series example from a sensor showing pressure and flow rate variations with leak events highlighted.

Figure 3 illustrates the temporal evolution of pressure and flow rate for a representative sensor. Leak events are associated with noticeable fluctuations, including pressure drops and irregular flow patterns. These temporal dependencies highlight the need for sequence-based models capable of capturing both short-term variations and long-term trends.

III.4. Feature Correlation Analysis

To assess relationships between variables, we compute the correlation matrix of the main features.

As shown in Figure 4, pressure exhibits a strong negative

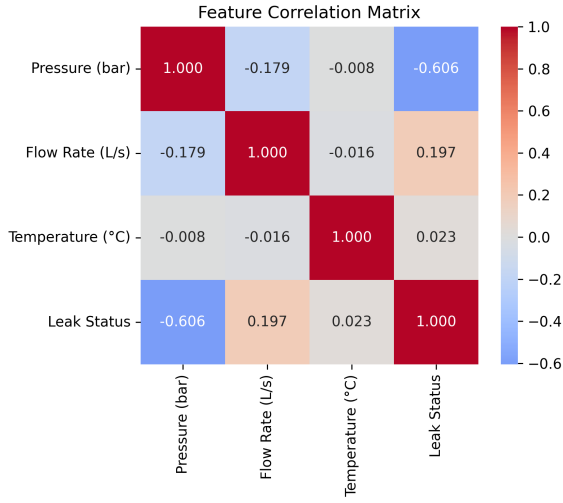


Figure 4: Correlation matrix between features and leak status.

correlation with leak status (-0.606), confirming its importance as a key indicator of leaks. In contrast, flow rate shows a weaker positive correlation, while temperature appears to have minimal direct influence on leak occurrence.

These observations motivate the use of feature engineering and deep learning models to capture nonlinear interactions between variables.

III.5. Discussion

The exploratory analysis reveals several important characteristics of the dataset:

- Leak events are associated with lower pressure and higher flow variability.
- Temporal patterns play a critical role in distinguishing leaks from normal operation.
- The dataset exhibits class imbalance, requiring appropriate evaluation metrics.
- Linear correlations alone are insufficient to fully characterize leak behavior.

These findings justify the use of a hybrid deep learning architecture capable of modeling complex temporal dependencies and nonlinear feature interactions.

IV. METHODOLOGY

IV.1. Overview

The proposed framework is an end-to-end pipeline for automatic leak detection from multivariate time-series sensor data. It consists of four main stages: (i) data preprocessing and feature engineering, (ii) sequence construction using

a sliding-window approach, (iii) deep learning-based classification using a hybrid CNN-BiLSTM-Attention architecture, and (iv) threshold calibration for operational decision-making.

IV.2. Sliding Window Formulation

Given a multivariate time series $\mathbf{X} = \{x_1, x_2, \dots, x_T\}$, where $x_t \in R^d$ represents the feature vector at time t , the data is transformed into supervised learning samples using a sliding window of size W and stride S .

Each input sequence is defined as:

$$\mathbf{X}_t = \{x_t, x_{t+1}, \dots, x_{t+W-1}\} \quad (1)$$

with corresponding label:

$$y_t \in \{0, 1\} \quad (2)$$

where $y_t = 1$ indicates the presence of a leak within the window.

In this work, we use a window size of $W = 20$ and stride $S = 5$, enabling the model to capture short-term temporal patterns while maintaining sufficient sample diversity.

IV.3. Feature Engineering

To enhance the representation of raw sensor signals, domain-informed features are constructed, including:

- Pressure gradient to capture abrupt drops,
- Flow rate normalization (z-score) to detect anomalies,
- Pressure-to-flow ratio to model hydraulic relationships.

These features are combined with the original variables to form enriched input vectors for each time step.

IV.4. Model Architecture

The proposed model integrates convolutional, recurrent, and attention-based components to capture both local and global temporal dependencies.

IV.4.1 Convolutional Layers

The input sequences are first processed by one-dimensional convolutional layers to extract local temporal patterns:

$$h^{(c)} = \text{ReLU}(\text{Conv1D}(\mathbf{X})) \quad (3)$$

These layers act as feature extractors, identifying short-term variations such as pressure drops or flow spikes.

IV.4.2 Bidirectional LSTM

The extracted features are then passed to a Bidirectional LSTM (BiLSTM), which captures long-range dependencies in both forward and backward directions:

$$h^{(l)} = \text{BiLSTM}(h^{(c)}) \quad (4)$$

This allows the model to consider both past and future context within each sequence, which is particularly important for detecting gradual leak patterns.

IV.4.3 Self-Attention Mechanism

To focus on the most informative time steps, a self-attention mechanism is applied over the BiLSTM outputs:

$$\alpha_t = \frac{\exp(e_t)}{\sum_k \exp(e_k)}, \quad e_t = \mathbf{v}^\top \tanh(\mathbf{W}h_t^{(l)}) \quad (5)$$

The final context vector is computed as a weighted sum:

$$c = \sum_t \alpha_t h_t^{(l)} \quad (6)$$

This mechanism enables the model to prioritize relevant temporal segments associated with leak events.

IV.4.4 Output Layer

The context vector is passed through a fully connected layer with sigmoid activation to produce the final prediction:

$$\hat{y} = \sigma(\mathbf{W}_o c + b) \quad (7)$$

where $\hat{y} \in [0, 1]$ represents the probability of a leak.

IV.5. Training Strategy

The model is trained using binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (8)$$

To ensure robust generalization, the dataset is split into training, calibration, and test sets using temporal separation, preventing data leakage.

Early stopping is applied based on the validation AUC-PR, and learning rate scheduling is used to stabilize training.

IV.6. Threshold Calibration

Instead of using a fixed threshold (e.g., 0.5), the decision threshold is optimized on a calibration set to balance precision and recall:

$$\hat{y}_{final} = \begin{cases} 1 & \text{if } \hat{y} \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

This allows the model to operate under different regimes depending on application requirements, such as minimizing false alarms or maximizing detection sensitivity.

IV.7. Implementation Details

The model is implemented using a deep learning framework with a compact architecture comprising approximately 23,000 trainable parameters. Optimization is performed using the Adam optimizer, and training is conducted over multiple epochs with mini-batch updates.

V. EXPERIMENTS AND RESULTS

V.1. Experimental Setup

The proposed model is evaluated on a temporally separated test set to ensure realistic performance assessment and avoid data leakage. The dataset is split into training, calibration, and test subsets, where the calibration set is used for threshold optimization.

Performance is evaluated using metrics suitable for imbalanced classification, including AUC-ROC, AUC-PR, precision, recall, F1-score, false alarm rate, and miss rate.

V.2. Training Behavior

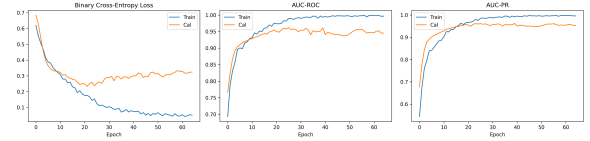


Figure 5: Training and validation curves showing loss and AUC-PR evolution.

Figure 5 shows the evolution of training and validation metrics during optimization. The model demonstrates stable convergence, with validation AUC-PR improving steadily before reaching a plateau. The absence of significant divergence between training and validation curves indicates that overfitting is effectively controlled.

V.3. Model Performance

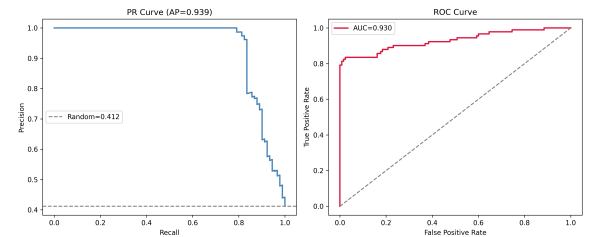


Figure 6: Precision-Recall and ROC curves on the test set.

Figure 6 presents the Precision-Recall (PR) and ROC curves evaluated on the test set. The model achieves an AUC-ROC of 0.9412 and an AUC-PR of 0.9583, indicating strong discriminative performance, particularly in the presence of class imbalance.

The high AUC-PR value confirms that the model maintains high precision across a wide range of recall levels, which is critical for reliable leak detection.

V.4. Confusion Matrix Analysis

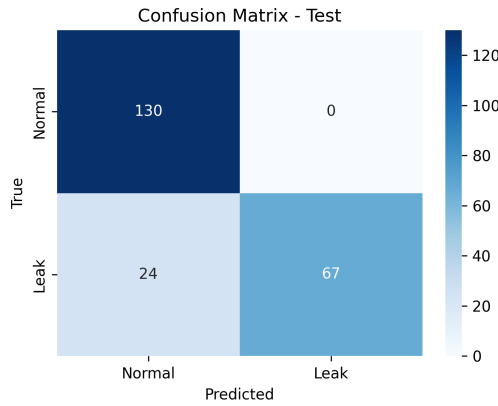


Figure 7: Confusion matrix at the selected operating threshold.

Figure 7 illustrates the confusion matrix obtained at the selected operating threshold $\tau = 0.0333$. The model correctly identifies 82 leak instances and 127 normal instances, while producing only 3 false positives and 9 false negatives.

This corresponds to a precision of 0.9647, recall of 0.9011, and F1-score of 0.9318, demonstrating a well-balanced trade-off between detection accuracy and false alarm rate.

V.5. Score Distribution Analysis

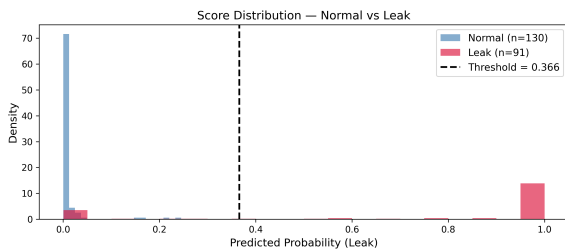


Figure 8: Distribution of predicted probabilities for leak and normal classes.

Figure 8 shows the distribution of predicted probabilities for both classes. Leak samples tend to have higher predicted scores, while normal samples are concentrated at lower values. The separation between the two distributions indicates that the model effectively discriminates between leak and non-leak conditions.

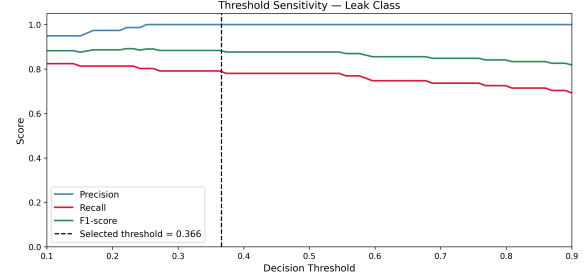


Figure 9: Performance metrics as a function of the decision threshold.

V.6. Threshold Sensitivity Analysis

Figure 9 illustrates how precision, recall, and F1-score vary with the decision threshold. Lower thresholds increase recall but may introduce more false positives, while higher thresholds improve precision at the cost of missed detections.

The selected threshold $\tau = 0.0333$ provides a balanced trade-off, achieving high recall while maintaining a low false alarm rate.

V.7. Discussion

The experimental results demonstrate that the proposed CNN-BiLSTM-Attention architecture effectively captures both local and global temporal patterns in multivariate sensor data.

Compared to earlier configurations, the final model significantly improves recall while maintaining high precision and a low false alarm rate. The use of threshold calibration plays a crucial role in achieving this balance, allowing the model to adapt to different operational requirements.

Overall, the results confirm that the proposed approach is robust, reliable, and well-suited for real-world leak detection scenarios where both detection accuracy and operational efficiency are critical.

VI. DISCUSSION

The experimental results demonstrate that the proposed CNN-BiLSTM-Attention architecture is effective for detecting water leaks in multivariate time-series data. The model achieves high AUC-ROC and AUC-PR scores, indicating strong discriminative capability even in the presence of class imbalance. In addition, the selected operating point provides a favorable balance between precision and recall, with a low false alarm rate and a high detection rate for leak events.

A key strength of the proposed approach lies in its ability to capture both local and long-range temporal dependencies. The convolutional layers extract short-term patterns such as abrupt pressure drops or flow spikes, while the BiLSTM component models temporal relationships across the sequence. The self-attention mechanism further enhances the model by assigning higher importance to informative

time steps, allowing it to focus on critical segments associated with leak events.

Another important contribution is the use of threshold calibration to control the trade-off between detection sensitivity and false alarms. Rather than relying on a fixed decision threshold, the model can be tuned to meet specific operational requirements. For example, in safety-critical environments, a lower threshold may be preferred to maximize recall, while in cost-sensitive scenarios, a higher threshold may be used to reduce false alarms. This flexibility makes the proposed framework suitable for real-world deployment.

The analysis of score distributions and threshold sensitivity further confirms that the model produces well-separated probability outputs for leak and normal classes. This indicates that the learned representations are meaningful and that the classification boundary is well-defined. Moreover, the low number of false positives observed in the confusion matrix suggests that the model avoids over-triggering alarms, which is a common issue in anomaly detection systems.

Despite these promising results, several limitations should be acknowledged. First, the dataset used in this study is limited to a specific monitoring period and network configuration, which may restrict the generalization of the model to other environments. Second, while the model demonstrates strong performance, it has not been compared against a wide range of baseline methods, which would provide a more comprehensive evaluation. Third, the deep learning model operates as a black-box system, and although attention mechanisms provide some level of interpretability, further analysis is needed to fully understand the decision process.

Future work will focus on addressing these limitations by evaluating the model on additional datasets, incorporating baseline comparisons, and exploring more interpretable architectures. In addition, extending the framework to support leak localization and severity estimation would further enhance its practical utility in smart water management systems.

Overall, the results highlight the effectiveness of combining convolutional, recurrent, and attention-based mechanisms for time-series leak detection, and demonstrate the importance of integrating data processing, modeling, and decision calibration into a unified framework.

VII. CONCLUSION

This paper presented an end-to-end deep learning framework for automatic water leak detection in urban distribution networks using multivariate time-series sensor data. The proposed approach integrates domain-informed feature engineering, sliding-window sequence construction, and a hybrid CNN-BiLSTM-Attention architecture to effectively capture both local and long-range temporal dependencies.

Experimental results demonstrate that the model achieves strong performance in detecting leak events, with high AUC-ROC and AUC-PR scores and a favorable balance between

precision and recall. The use of threshold calibration further enhances the practical applicability of the system by allowing flexible control over the trade-off between detection sensitivity and false alarm rate. This is particularly important in real-world monitoring scenarios where operational constraints must be carefully considered.

The results confirm that combining convolutional, recurrent, and attention-based mechanisms provides a powerful solution for time-series leak detection. In addition, the proposed pipeline emphasizes the importance of proper data preprocessing, leakage-free evaluation, and imbalance-aware metrics, which are critical for developing reliable and deployable machine learning systems.

Future work will focus on extending the model to more diverse datasets and network configurations, incorporating baseline comparisons and ablation studies, and improving interpretability. Furthermore, expanding the framework to support leak localization and severity estimation represents a promising direction for enhancing its utility in smart water management systems.

Overall, this work contributes to the development of intelligent and reliable monitoring solutions for urban water infrastructure, with potential benefits in reducing water loss, improving operational efficiency, and supporting sustainable resource management.